

Programming Assignment 4.1

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Replicated state machine (GigaPaxos): GigaPaxos to encapsulate the application as an RSM

We know GigaPaxos ensures every server receives the same requests in the exact same order.

In this design, gigapaxos is basically a consensus engine which follows the replication logic.

Code Design (method definitions):

- **execute():**
Inputs request from gigapaxos that has been ordered by the cluster. Treat the request string as raw CQL command and execute it in local db. It handles the inserts and updates to run perfectly.
It also enables replay the missed commands by calling execute continuously.
- **checkpoint()**
This ensures to handle if the internal logs of request become too big(max log size is 400). This is basically a snapshotting mechanism, it runs select * on the entire current state of db. We convert every row into json string and return this to gigapaxos which helps it to clear logs.
- **restore()**
This enables state transfer that is when a server is dead for a long time and it misses requests for which logs have been deleted and one cannot replay the requests. Hence, we essentially truncate the current database table and then rebuild.

Flow of code:

- 1) Constructor basically connects to the DB.
- 2) Running will be handled by execute method to run the CQL commands.
- 3) Logging will be handled by checkpoint that dumps db to json
- 4) In cases of crash or restart:
 - a) If the downtime is short: execute will replay the missed commands
 - b) If the downtime is long: restore will wipe the DB and load from JSON.